

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.3 WAVES – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How do waves carry energy through our environment, and how can understanding their behaviour protect Kenyan communities, improve technology, and help us make sense of the world around us? In this task you will act as a scientist explaining the Lake Victoria fisherman's puzzle. Write your answers in the space under each prompt. Use the vocabulary and ideas from Lessons 1 to 8: wavelength, frequency, wave speed, reflection, refraction, diffraction, and the wave equation. Show every step of any calculation, including units. Refer back to the fisherman near Kisumu in each answer. Write in full sentences, and label any diagram you draw.

Part 1: The wave equation and the fisherman's sounds and waves

A fisherman near Kisumu hears a deep thunder rumble and sees water waves crossing the lake.

(a) Write the wave equation and say what each symbol means. (b) A Lake Victoria boat-engine wave has wavelength 0.8 m and speed 3.2 m/s. Calculate its frequency and its period. (c) Explain why, in the same medium, a wave of higher frequency has a shorter wavelength. (d) Say how the wave equation helps you describe the low-frequency thunder rumble compared with a high-pitched sound.

Write your answer here:

Part 2: Reflection – what the echo can and cannot explain

In Lesson 3 you clapped at measured distances from the school perimeter wall and timed the echo.

(a) State the law of reflection. (b) Explain what an echo is. (c) The echo-distance formula is $d = (v \times t) / 2$. Explain why the time is divided by 2. (d) A sonar pulse from a boat in Kisumu harbour travels at 1500 m/s and returns after 0.2 s. Calculate the depth of the water. (e) Reflection can explain part of what the fisherman hears. Explain what it can explain and what it cannot explain about the sound reaching the cove behind the headland.

Write your answer here:

Part 3: Refraction of the water waves

The water waves from the storm change direction as they cross from the deep open lake into the shallow bay near the Kisumu shore.

(a) Define refraction. (b) Explain why the wavefronts change direction at the boundary between deep and shallow water, and say which way they bend relative to the normal. (c) A wave has frequency 680 Hz. It travels at 340 m/s in one region of air and slows to 320 m/s in another. Calculate the wavelength in each region. (d) What stays the same and what changes for a wave crossing a boundary? (e) Describe one Kenyan example of refraction of sound that you studied.

Write your answer here:

Part 4: Diffraction and putting the whole phenomenon together

(a) Define diffraction and say when it is greatest. (b) Use diffraction to explain how the deep thunder rumble reaches the fisherman even when the storm is behind the headland. (c) The mobile-phone signal fades when you go behind a hill in the Rift Valley. Explain what is happening and what engineers do about it. (d) Write a short explanation of the full fisherman phenomenon, naming which wave behaviour (reflection, refraction or diffraction) explains each of the three observations: the echo from the cliff, the bending water waves at the bay, and the sound in the cove.

Write your answer here:

Part 5: Applications, environment and the Scientist's Statement

(a) Choose TWO Kenyan applications of wave properties from Lesson 6 (for example KMFRI sonar on Lake Victoria, ultrasound at Kenyatta National Hospital, cats-eye reflectors, or seismic monitoring by the Kenya Meteorological Department). For each, name the wave property it uses and explain how. (b) Describe ONE harmful effect of waves on the Kenyan environment and the wave property that causes it. (c) Describe ONE mitigation measure and explain the wave physics of how it works. (d) Write a Scientist's Statement of 4 to 6 sentences to the Kisumu fisherman explaining how wave science helps keep coastal communities safer.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of the wave equation and calculations	All calculations are set out correctly with formulas, steps and units (4 Hz, 0.25 s, 150 m, 0.50 m and 0.47 m), and the inverse relationship between f and λ is clearly explained.	Most calculations are correct with some units or steps missing; the f and λ relationship is stated but not fully explained.	Calculations contain errors or are missing; the wave equation is stated without being applied correctly.
Explanation of reflection, refraction and diffraction	Each behaviour is defined accurately and the mechanism is explained (angle law; speed change at a boundary with bending towards the normal and constant frequency; diffraction greatest when size is close to wavelength).	Each behaviour is defined correctly but one mechanism is incomplete or imprecise.	Behaviours are confused with each other or described without any mechanism.
Connection to the anchoring	Each of the three observations (cliff echo,	Observations are matched to wave	Few links are made to the fisherman at

phenomenon	bending water waves, sound in the cove) is matched to the correct wave behaviour, and the limits of reflection and refraction are explained.	behaviours with minor gaps in the reasoning.	Kisumu, or the matches are incorrect.
Kenyan applications and environmental impact	Applications, impacts and mitigation are accurate, use Kenyan contexts, and name the wave property and the physics of how it works.	Applications and impacts are correct but the physics of how they work is only partly explained.	Examples are vague, not linked to wave properties, or incorrect.
Scientific communication and Scientist's Statement	Writing is clear, uses correct vocabulary, and the statement to the fisherman is accurate, plain-language and links wave science to community safety.	Writing is mostly clear with some vocabulary slips; the statement is accurate but links to safety are brief.	Writing is unclear or vocabulary is misused; the statement is missing or inaccurate.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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